SIADS 696 Milestone II – Team 17

Healthcare Access Characteristics:
Predicting Routine Checkup Rates and Identifying County Access – Vulnerability Profiles

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1. Introduction

Preventive healthcare utilization varies across U.S. communities and is associated with social vulnerability, insurance coverage, rurality, and healthcare access. Prior studies have documented disparities in preventive-care use and shown that counties can exhibit different combinations of social and healthcare-resource characteristics rather than falling along a single continuum of advantage or disadvantage (Bauer et al., 2022; Al Rifai et al., 2022; Bowser et al., 2024).

This project integrates four national county-level data sources—CDC PLACES, the CDC/ATSDR Social Vulnerability Index (SVI), the Health Resources and Services Administration (HRSA) Area Health Resources Files (AHRF), and HRSA Health Professional Shortage Area (HPSA) data—to address two questions:

Supervised research question: How well can county-level healthcare-resource and access characteristics predict the age-adjusted prevalence of adults receiving a routine checkup within the past year?

Unsupervised research question: What healthcare-access and social-vulnerability profiles emerge among U.S. counties, and do those independently identified profiles differ in routine-checkup utilization?

For supervised learning, we compare Elastic Net Regression, Support Vector Regression (SVR), and Gradient Boosting Regression against a median DummyRegressor baseline. The primary feature set emphasizes healthcare resources and access rather than full SVI demographics because PLACES estimates are themselves produced using BRFSS, Census, and demographic inputs; including strongly overlapping predictors could introduce shared-input circularity.

For unsupervised learning, K-Means and Gaussian Mixture Models (GMM) are used to identify county profiles from healthcare-resource and social-vulnerability characteristics. Routine-checkup prevalence is excluded from cluster formation and reintroduced only afterward for descriptive comparison.

SVR achieved the best primary supervised cross-validation performance, with a mean MAE of 2.105 percentage points, 25.2% lower than the Dummy baseline. Gradient Boosting performed nearly as well, while performance declined under state-grouped validation, indicating weaker generalization to entirely unseen states. In the unsupervised analysis, five interpretable county profiles emerged, but substantial overlap showed that counties lie along a continuum rather than forming sharply separated types.

2. Related Work

Bauer et al. (2022) examined county-level social vulnerability and cancer-screening rates across 3,141 U.S. counties using CDC PLACES and SVI. Our project extends this framework to routine preventive checkups and incorporates a broader set of healthcare-workforce and infrastructure measures from AHRF and HPSA.

Al Rifai et al. (2022) studied state-level SVI and healthcare access among adults with cardiovascular disease, including insurance, primary-care access, and routine checkups. Our analysis differs by focusing on county-level population characteristics and directly integrating local healthcare-resource availability.

Bowser et al. (2024) used more than 650 variables from 24 datasets and factor analysis with K-Means to identify broad U.S. county types. Our clustering uses a smaller, deliberately interpretable feature set, compares K-Means with probabilistic GMM clustering, and externally examines routine-checkup utilization after clusters are formed.

The project builds conceptually on prior team work involving healthcare access, social vulnerability, and chronic-disease indicators. Samantha previously conducted a California-focused healthcare-access analysis involving AHRF, HPSA, and SVI data; Gina previously examined SVI in relation to diabetes outcomes; and Sara previously analyzed chronic-disease indicators and comorbidities using CDC and Census data. However, the present Milestone II project uses a newly constructed national county-level analytical dataset, new research questions, and newly developed supervised and unsupervised learning workflows. **No previously graded supervised or unsupervised learning code was reused for this project.** Any overlap with prior work is limited to domain knowledge, familiarity with the public data sources, and general data-preparation concepts rather than previously graded machine-learning analyses.

3. Data Sources

The project integrates four national public data sources: CDC PLACES, the HRSA Area Health Resources Files (AHRF), HRSA Health Professional Shortage Area (HPSA) records, and the CDC/ATSDR Social Vulnerability Index (SVI). PLACES, HPSA, and SVI were obtained and processed as CSV files, while AHRF was obtained as the HRSA 2023–2024 compressed data archive. Standardized five-character county FIPS codes were used to harmonize the sources. CDC PLACES defined the project geographic universe, yielding 3,144 U.S. counties and county equivalents after removal of the national aggregate. The final integrated analytical dataset contains one observation per county/county equivalent and 50 retained variables.

3.1 CDC PLACES

CDC PLACES provides model-based small-area estimates of chronic disease, health behaviors, disability, and preventive-service utilization. The project uses the 2024 county release and focuses on the 2022 age-adjusted prevalence of adults reporting a routine medical checkup within the past year (`routine_checkup_pct`).

This measure serves as the supervised-learning outcome. PLACES estimates are derived from BRFSS data together with Census and American Community Survey information, which is important for feature-selection decisions later in the analysis. In particular, the full set of SVI demographic variables was not included in the primary supervised specification because of potential conceptual overlap with information used to construct PLACES estimates.

After excluding the U.S. aggregate, the PLACES data contained 3,144 unique county/county-equivalent FIPS codes. These records defined the authoritative geographic universe used for subsequent data integration.

3.2 HRSA Area Health Resources Files

The 2023–2024 HRSA Area Health Resources Files provide county-level information on healthcare workforce, facilities, population, rurality, and healthcare-system characteristics. The full geographic extract contained 3,240 unique geographic records. All 3,144 PLACES counties/county equivalents were represented; the remaining 96 records corresponded to territorial, historical, or superseded geographies outside the project's analytical universe.

Selected AHRF variables were chosen to represent healthcare-resource availability and geographic context. These included counts of primary-care physicians, nurse practitioners, physician assistants, hospitals, hospital beds, Federally Qualified Health Centers, and Rural Health Clinics, along with population density and Rural-Urban Continuum Codes. Provider and facility counts were later converted to population-standardized rates to improve comparability across counties of different sizes.

Although the project targets approximately 2022 conditions, individual AHRF variables originate from different underlying reference years. This temporal variation is retained as a limitation of the analysis rather than treating the 2023–2024 AHRF release as representing a single measurement year.

3.3 HRSA Health Professional Shortage Area Data

HRSA HPSA data provide information about designated shortages in Primary Care, Mental Health, and Dental Health services. Unlike PLACES, SVI, and the retained AHRF records, the source data are designation-level rather than one-row-per-county. A single designation may also contain multiple county components.

To align HPSA information with the approximately 2022 analytical period, designation and withdrawal dates were used to reconstruct whether qualifying geographic or population HPSA designations were plausibly active on December 31, 2022. Eligible records were then aggregated to the county level. Derived measures included designation presence, the number of qualifying designations, and maximum recorded HPSA score for the three retained disciplines.

Because the analysis relies on a reconstruction from current HRSA extracts rather than an official archived 2022 HPSA snapshot, binary designation presence was treated as the most defensible primary shortage measure. Counts and scores were used primarily in sensitivity analyses. Facility-specific HPSAs were excluded because facility-level shortage status does not necessarily characterize access conditions for the broader county population.

3.4 CDC/ATSDR Social Vulnerability Index

The 2022 CDC/ATSDR Social Vulnerability Index characterizes community-level social conditions that may affect a population's ability to prepare for, respond to, and recover from adverse events. The SVI data provided exact FIPS coverage for all 3,144 counties/county equivalents in the PLACES-defined project universe.

The unsupervised analysis uses the 16 underlying SVI percentage indicators, representing socioeconomic status, household characteristics, racial and ethnic minority status and language, and housing and transportation conditions. Examples include poverty, unemployment, lack of health insurance, disability, limited English proficiency, crowding, lack of vehicle access, and housing characteristics.

The component indicators were retained rather than simultaneously including the derived SVI theme and overall percentile rankings, which would introduce redundant representations of the same underlying constructs. SVI values encoded as −999 were interpreted as missing rather than valid extreme observations. The 16 indicators form the primary social-vulnerability representation for clustering and are evaluated separately as an expanded feature set in the supervised sensitivity analysis.

3.5 Integrated Analytical Dataset

The four sources were harmonized using standardized five-character county FIPS codes. County names were retained for interpretation and quality-control purposes but were not used as merge keys.

Incremental joins were validated as one-to-one so that counties were neither lost nor duplicated during integration.

The resulting shared analytical dataset contains 3,144 county/county-equivalent observations and 50 retained variables. PLACES provides the supervised outcome; AHRF provides healthcare-resource, infrastructure, rurality, and population-density measures; HPSA provides provider-shortage indicators; and SVI provides the primary social-vulnerability representation used in the unsupervised analysis.

Nine Connecticut planning regions contain structured missingness in several historical AHRF healthcare-resource variables following the state's transition from counties to planning regions. These observations remain in the shared dataset and supervised analysis but are excluded from the primary clustering matrix, producing an unsupervised sample of 3,135 observations. Complete feature definitions, source assignments, analytical roles, geographic harmonization results, and missing-data exceptions are reported in **Appendix Tables A1–A3**.

4. Data Preparation and Feature Engineering

PLACES defined the project universe of 3,144 counties/county equivalents. All sources were harmonized by FIPS and merged with validation checks to ensure no counties were duplicated or lost.

AHRF provider and facility counts were converted to population-standardized rates. Engineered measures included primary-care physicians, nurse practitioners, physician assistants, hospitals, FQHCs, and Rural Health Clinics per 100,000 residents; hospital beds per 1,000 residents; log population density; and a simplified metro/nonmetro classification.

HPSA records were aggregated to county-level designation presence, count, and maximum score for Primary Care, Mental Health, and Dental Health. SVI sentinel values of −999 were converted to missing values.

No global statistical imputation was applied before modeling. In the supervised workflow, imputation, scaling, transformation, and encoding were performed inside model pipelines to avoid leakage. For clustering, transformations, imputation, and standardization were performed before model fitting.

Nine Connecticut planning regions had structured missingness in historical AHRF variables because of the state's transition from counties to planning regions. They were retained in supervised analyses with pipeline imputation but excluded from the primary clustering matrix. The complete feature definitions and analytical roles are provided in **Appendix Table A1**.

5. Part A: Supervised Learning

5.1 Feature Representation and Modeling Workflow

The primary supervised specification contained 10 predictors spanning four domains:

- **Provider supply:** primary-care physicians, nurse practitioners, and physician assistants per 100,000.
- **Healthcare infrastructure:** hospitals, FQHCs, and Rural Health Clinics per 100,000 and hospital beds per 1,000.
- **Geographic/access context:** log population density and metro/nonmetro status.
- **Provider shortage:** primary-care HPSA designation presence.

The full SVI feature set was excluded from the primary specification because of potential shared-input overlap with PLACES estimation and was instead evaluated as a sensitivity analysis.

A median DummyRegressor provided the baseline. Elastic Net represented a regularized linear approach; SVR provided a kernel-based nonlinear model; and Gradient Boosting represented a nonlinear tree ensemble. Model-specific preprocessing was contained within scikit-learn pipelines so that imputation, scaling, and encoding parameters were estimated using only the relevant training data.

Hyperparameters were tuned separately for each substantive model family using cross-validation within the training data, with MAE used as the model-selection and refit criterion. The search included regularization strength and mixing parameters for Elastic Net; C, epsilon, gamma, and feature-rate representation for SVR; and learning rate, number of trees, tree depth, and loss function, with minimum leaf size and subsampling fixed for Gradient Boosting. The complete hyperparameter search spaces and selected configurations are reported in **Appendix Table B1**.

A holdout test set was reserved before model selection and was not used during hyperparameter tuning. Models were compared using five-fold random cross-validation, with state-grouped cross-validation used as a geographic robustness analysis. The reserved holdout set was evaluated only after the final model-selection decision had been made.

5.2 Evaluation and Model Comparison

MAE was the primary metric because it measures prediction error directly in percentage points**. RMSE** was used to emphasize larger errors**,** and **R^2** summarized explained variation.

Table 1. Supervised Model Comparison

Model	Random-CV MAE	Random-CV RMSE	Random-CV R^2	State-grouped MAE
Dummy	2.813 ± 0.064	3.631 ± 0.063	−0.027 ± 0.009	3.056 ± 0.744
Elastic Net	2.329 ± 0.052	2.999 ± 0.078	0.299 ± 0.034	2.569 ± 0.448
SVR	**2.105 ± 0.027**	**2.756 ± 0.035**	**0.408 ± 0.018**	**2.455 ± 0.562**
Gradient Boosting	2.116 ± 0.064	2.776 ± 0.054	0.399 ± 0.019	2.461 ± 0.597

Note: Values are cross-validation mean ± fold standard deviation. Lower MAE/RMSE and higher R^2 indicate better performance.

After model selection was finalized, the reserved holdout test set was evaluated as an independent confirmation step. The selected SVR achieved a holdout MAE of **2.053 percentage points**, RMSE of **2.696**, and R^2 of **0.427**. Its holdout MAE was close to its random-CV estimate of 2.105, providing evidence that the cross-validation estimate generalized reasonably well to previously unseen observations. The SVR holdout MAE had a 95% bootstrap confidence interval of **1.921–2.186 percentage points**. The holdout set was used only for final evaluation of the selected SVR and was not used for model selection or hyperparameter tuning.

5.3 Feature Importance and Ablation

Permutation importance for the selected SVR showed that **log population density was by far the strongest individual predictor**, followed by substantially smaller contributions from hospitals and provider-supply measures. Primary-care HPSA designation contributed little predictive information in the primary specification.

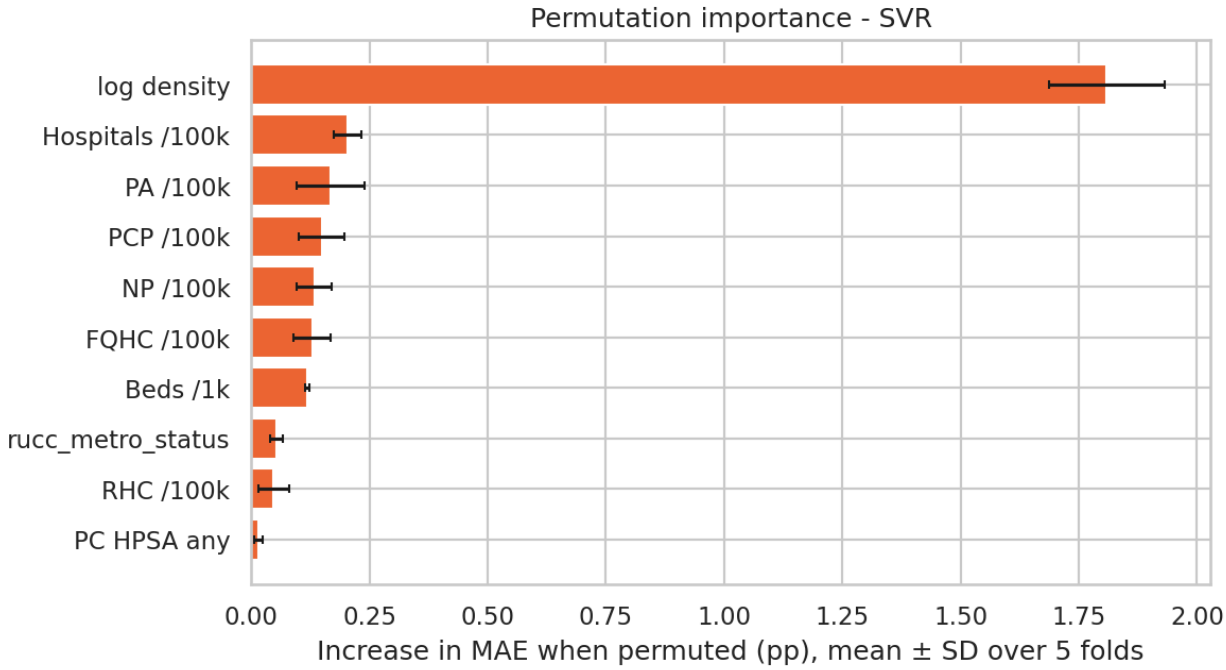


Figure 1. Permutation feature importance for the selected SVR model. Importance is measured as the increase in cross-validated MAE after each feature is permuted; larger increases indicate greater predictive importance.

Feature-group ablation evaluated the unique contribution of broader domains.

Table 2. Feature-Group Ablation for SVR

Feature group removed	Random-CV MAE	ΔMAE vs. full model
Geographic/access context	2.441	+0.336
Healthcare infrastructure	2.229	+0.124
Provider supply	2.158	+0.053
Provider shortage	2.105	~0.000
Full SVR	**2.105**	—

Note: Removing geographic/access context caused the largest deterioration in performance. Healthcare infrastructure and provider supply added smaller amounts of unique predictive information, while removing primary-care HPSA designation produced essentially no loss in random-CV performance.

5.4 Sensitivity Analysis

Alternative specifications evaluated HPSA counts/scores, detailed rurality, raw rather than log population density, and addition of 16 SVI indicators. Each specification was evaluated with the selected SVR and with Elastic Net, the interpretable linear reference, to check whether conclusions about feature representation depended on the model family.

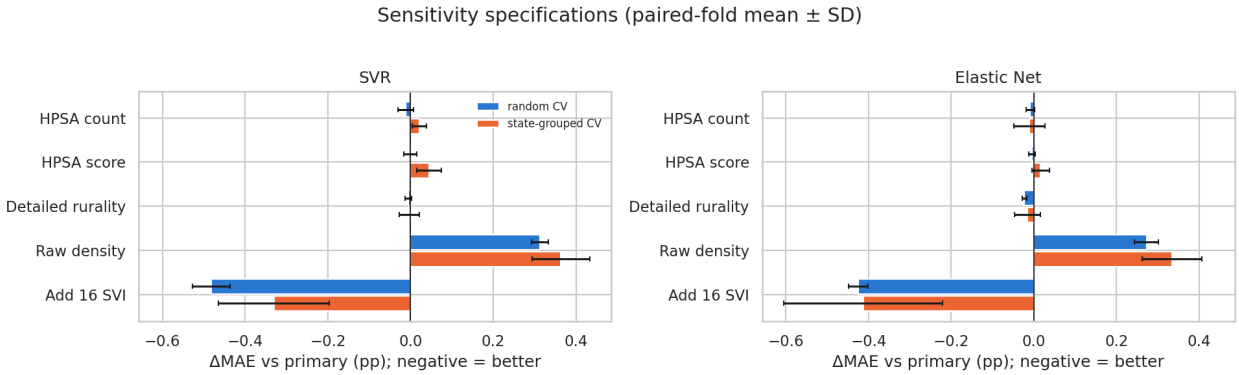


Figure 2. Sensitivity of SVR and Elastic Net to alternative feature specifications. Bars show the change in MAE relative to the primary specification under random and state-grouped cross-validation; negative values indicate improved performance.

Most alternative HPSA and rurality representations produced only small changes. Replacing log population density with raw density substantially worsened SVR performance, while adding SVI produced the largest improvement, reducing random-CV MAE from 2.105 to 1.624. Because SVI overlaps conceptually with demographic information used in PLACES estimation, however, the healthcare-resource specification was retained as the primary model.

The analysis therefore revealed three important tradeoffs. First, SVR improved accuracy relative to Elastic Net but was less directly interpretable. Second, adding SVI improved accuracy but reduced conceptual separation from the PLACES estimation process. Third, SVR MAE increased from 2.105 under random CV to 2.455 under state-grouped CV, showing that generalization to entirely unseen states was more difficult.

5.5 Failure Analysis

The selected SVR showed a clear tendency to predict values toward the middle of the outcome distribution. Counties with unusually low routine-checkup prevalence were often overpredicted, while counties with unusually high prevalence were often underpredicted. Three substantial errors illustrate different situations in which the model struggled (Table 3).

Table 3. Examples of substantial SVR prediction errors.

County	Observed Checkup (%)	Predicted (%)	Residual (Observed-Predicted)	Failure Pattern
Kalawao, HI	59.3	71.60	**-12.30**	Very small county; model substantially overpredicted utilization
Denver, CO	66.0	76.00	**-10.00**	Large metropolitan county with relatively high resource availability and density but lower-than-predicted utilization.
Piscataquis, ME	78.9	69.05	**+9.85**	Nonmetropolitan county with substantially higher utilization than predicted

These cases demonstrate that resource availability alone does not capture all factors associated with preventive-care use. Future work could incorporate additional policy, insurance, healthcare-system, behavioral, or regional variables; use hierarchical or spatial modeling; and consider approaches more robust to very small county populations.

6. Part B: Unsupervised Learning

6.1 Feature Representation and Methods

The primary clustering sample contains **3,135 counties/county equivalents** after excluding nine Connecticut planning regions with structured AHRF missingness. The clustering matrix contains 24 standardized continuous features: eight healthcare-resource/context measures and 16 SVI indicators. Routine-checkup prevalence was not used to construct clusters.

K-Means was selected as an interpretable distance-based approach whose centroids can be translated directly into county profiles. GMM provided a probabilistic alternative that permits components with different shapes, sizes, and covariance structures and assigns membership probabilities rather than only hard labels.

6.2 K-Means Results

Candidate values of (k=2) through (10) were compared using silhouette score, Davies–Bouldin index, Calinski–Harabasz index, stability, cluster size, and interpretability. Although (k=2) produced the highest silhouette score, it represented only a broad division. The five-cluster solution reached the lowest Davies–Bouldin score, a local silhouette maximum, and very high stability across seeds (mean ARI = 0.992; minimum = 0.984).

Silhouette scores were low overall, indicating that clusters should be interpreted as **profiles along a continuum rather than sharply separated county types**.

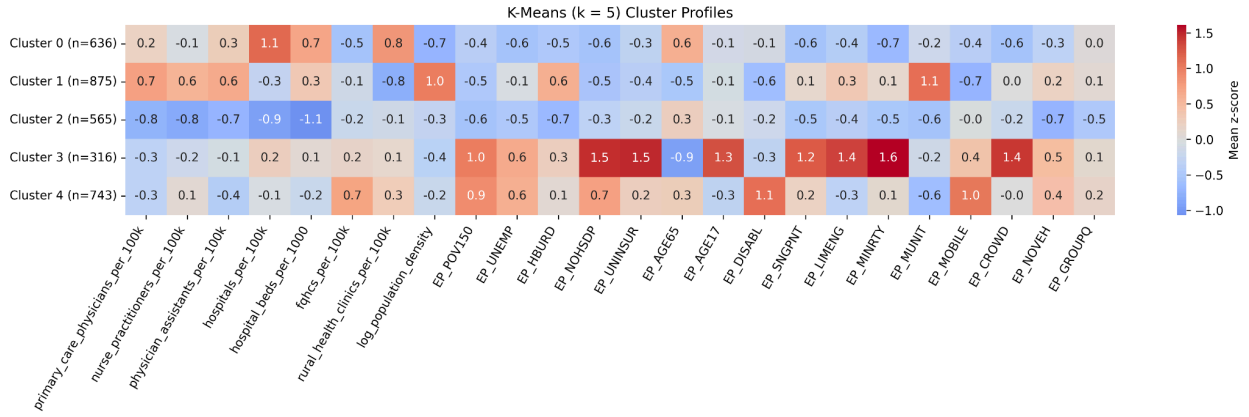


Figure 3. K-Means cluster profiles. Cells show mean standardized feature values within each of the five county profiles.

The five K-Means profiles were:

Cluster	Profile	Key characteristics
0 (n=636)	Rural, older, hospital-anchored	More hospitals/beds and RHCs; low density; older population; relatively low vulnerability
1 (n=875)	Urban/suburban, provider-rich	High density and multi-unit housing; highest provider supply; relatively low poverty/disability
2 (n=565)	Resource-scarce, low-vulnerability	Low provider and hospital supply despite comparatively low social vulnerability
3 (n=316)	High access barriers, young	High minority share, uninsurance, limited English, crowding, and poverty; younger population
4 (n=743)	Rural high-hardship, FQHC-reliant	High disability and poverty; greater FQHC reliance; less urban housing

A notable finding was that **healthcare-resource scarcity and social vulnerability were not the same dimension**. Cluster 2 had the lowest provider and hospital capacity but relatively low social vulnerability, while Clusters 3 and 4 showed substantial vulnerability with different resource profiles.

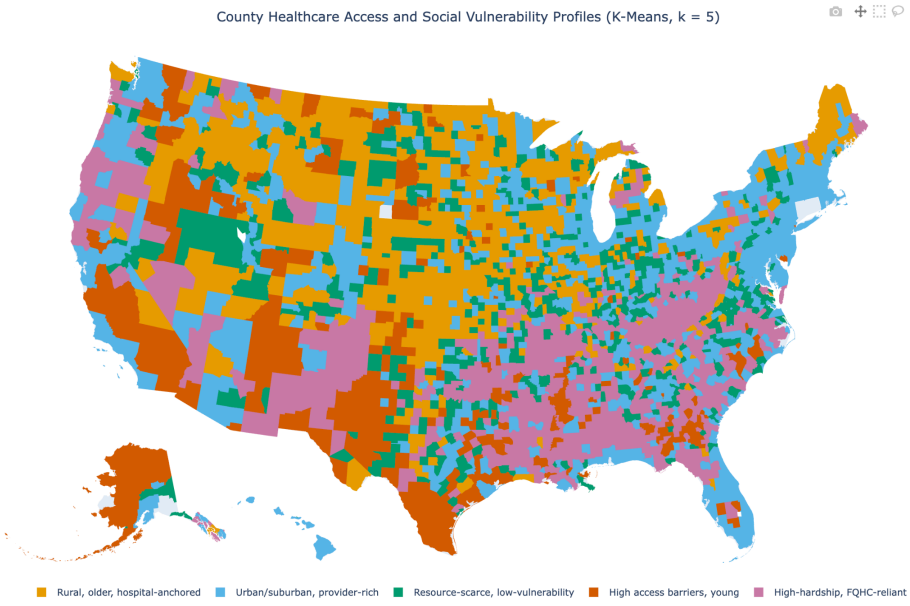


Figure 4. K-Means county choropleth. Geographic distribution of the five K-Means healthcare-access and vulnerability profiles.

6.3 Gaussian Mixture Models

GMM specifications varied covariance structure and number of components and were evaluated using BIC/AIC, stability, component sizes, separation, and interpretability. Full-covariance models fit substantially better than spherical, diagonal, or tied alternatives.

The nine-component model achieved the lowest BIC but was considerably less stable across random seeds (mean ARI = 0.747; minimum = 0.578). The five-component full-covariance solution was much more stable (mean ARI = 0.955; minimum = 0.921) and easier to interpret. It was therefore retained as the primary GMM comparison, with (k=9) treated as exploratory evidence of finer substructure.

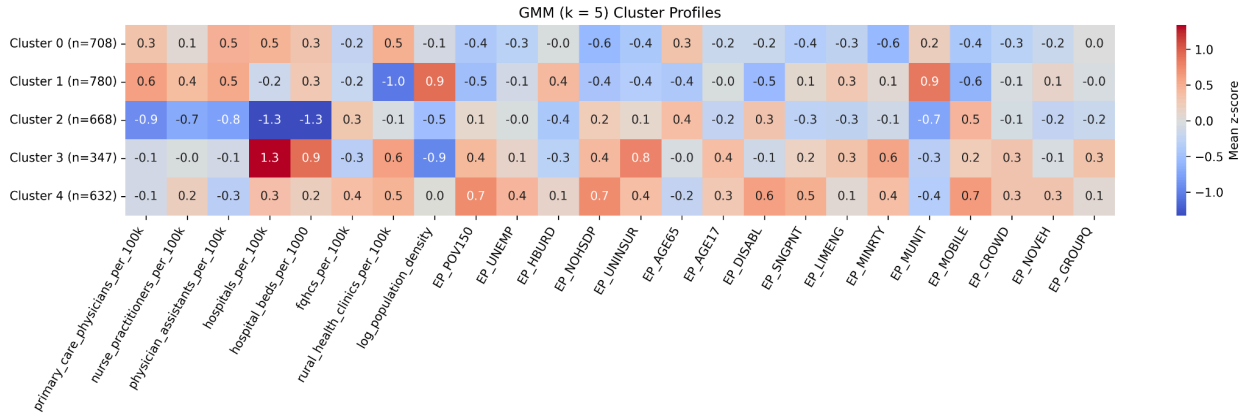


Figure 5. GMM cluster-profile heatmap. GMM five-component cluster profiles based on mean standardized feature values.

6.4 Comparison of K-Means and GMM

Table 4. Unsupervised Model Comparison

Evaluation criterion	K-Means (k = 5)	GMM (full, k = 5)
Silhouette score	0.123	0.023
Mean stability ARI	0.992	0.955
Minimum stability ARI	0.984	0.921
Clusters with defining feature ($\|z\| \geq 1.0$)	5 of 5	3 of 5
Final role	**Primary model**	**Comparison model**

Note: Models are compared on separation, stability, and interpretability.

K-Means and GMM showed only moderate overall agreement (ARI = 0.29), but both recovered several broad structures, including urban/provider-rich, resource-scarce, and rural/hospital-oriented profiles. The

greatest disagreement occurred among high-vulnerability counties, indicating that those finer boundaries were method-dependent.

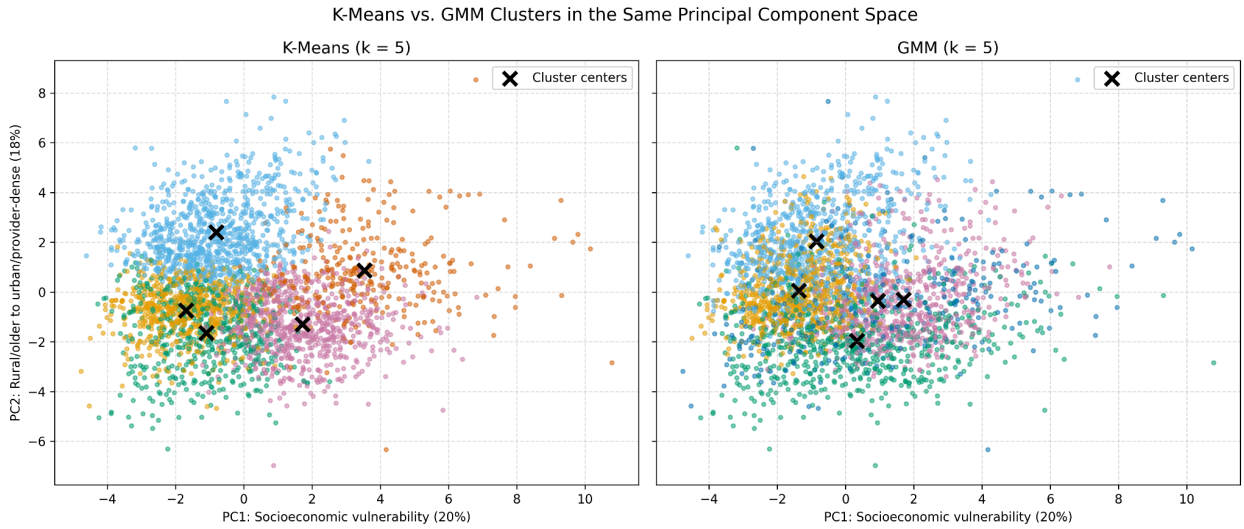


Figure 6. K-Means and GMM projected in the same PCA space. Both methods recover broad county structure, while boundaries among overlapping vulnerability profiles differ.

K-Means was selected as the primary unsupervised model because it was more stable, produced clearer defining features, and provided more interpretable profiles.

6.5 Sensitivity Analysis

Sensitivity analyses evaluated random-seed stability, alternative feature weighting, and addition of HPSA designation flags.

Table 5. Unsupervised Sensitivity Analyses

Check	Model	ARI vs. primary	Interpretation
Seed stability (5 reruns)	K-Means k = 5	0.992	Highly stable
Seed stability (5 reruns)	GMM k = 5	0.955	Highly stable
Seed stability (5 reruns)	GMM k = 9	0.747	Moderately stable; treated as exploratory
Add HPSA flags	K-Means k = 5	0.403	Boundaries shift; clusters stay balanced
Add HPSA flags	GMM k = 5	0.028	Degenerate (one component with 62% counties)
Equal domain weighting	K-Means k = 5	0.364	Broad structure kept; vulnerability subtypes merge

Note: Equal weighting of healthcare-resource and SVI domains materially changed finer cluster assignments, particularly by merging the two high-vulnerability profiles. Adding HPSA flags also changed assignments substantially, especially for GMM. These findings suggest that broad access/vulnerability structure is recognizable, but finer cluster boundaries depend on feature representation.

HPSA designation was therefore retained as a complementary profiling and sensitivity measure rather than part of the primary continuous clustering matrix.

6.6 Routine-Checkup Utilization Across Profiles

After clusters were finalized independently, PLACES routine-checkup prevalence was reintroduced for descriptive comparison.

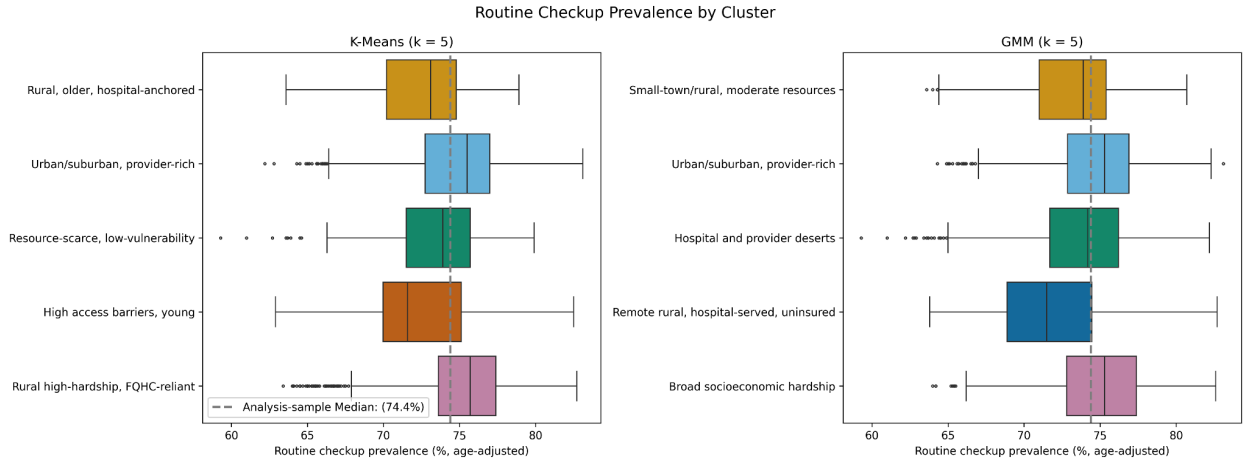


Figure 7. Routine-checkup prevalence across independently identified county profiles. Distributions overlap substantially, but the lowest median prevalence occurs in profiles characterized by high uninsurance and related access barriers.

Median routine-checkup prevalence ranged from approximately 71.5% to 75.7%, compared with an overall clustering-sample median of 74.4%. The lowest values occurred in profiles characterized by high uninsurance and access barriers rather than in the profile defined primarily by provider scarcity. Because distributions overlap considerably, these differences are descriptive and should not be interpreted causally.

7. Discussion

7.1 Supervised Learning Findings

The supervised analysis showed that county-level healthcare-resource and access characteristics can predict routine-checkup prevalence meaningfully better than a simple baseline, but they explain only part of the variation across counties. The selected SVR reduced prediction error by about 25% relative to the median baseline. Its near-tie with Gradient Boosting, together with Elastic Net's weaker performance, indicated that the relationships between access characteristics and utilization are nonlinear. One of the most surprising findings was how little the more specific access measures contributed. Geographic context, especially population density, carried most of the predictive signal. Provider supply and facility infrastructure added smaller amounts, and primary-care HPSA designation added essentially nothing in any representation.

A central challenge was that routine-checkup utilization is strongly structured by state. State membership alone accounted for about 88% of the variation in the outcome, and much of the model's remaining error was shared by counties within the same state. To test how much this mattered, we compared random cross-validation with state-grouped cross-validation for all model families and examined state-level residual bias. Under state-grouped cross-validation, error increased and R^2 became negative for every

model. The selected SVR still predicted with lower error than the median baseline, but it could not recover the overall checkup level of a state it had not seen. Because the holdout test set was drawn randomly across counties, the selected model's test result reflects performance within states represented in training rather than generalization to new states. We treat this as a substantive limitation rather than a technical detail: the patterns the model learns are reliable within the states represented in training, but much less transferable to new state contexts.

The sensitivity analyses also showed that predictive performance and conceptual validity can pull in different directions. Adding SVI produced the largest improvement in accuracy and also changed the interpretation. Hospital and physician supply lost most of their importance, while density and nonphysician provider supply remained relatively important. This suggests that some apparent resource effects partly reflected the composition of the communities where those resources are located. Because several SVI indicators overlap with information used to construct PLACES estimates, we retained the access-only model as the primary specification and treated the SVI model as an upper bound. More broadly, feature importance describes what the model relies on for prediction, not what causes routine-checkup utilization. The tendency to compress predictions toward the middle of the distribution further indicates that the most unusual counties differ in ways that county resource measures do not capture.

With additional time and resources, the analysis could be extended in ways that target these failure patterns directly. Hierarchical or spatial models with explicit state-level covariates, such as insurance coverage and Medicaid policy, could help explain the systematic state-level offsets and improve generalization to unseen states. Smoothing or population weighting could reduce the influence of very small counties whose rates and model-based estimates are unstable. Combined with the unsupervised results, which point to the same conclusion, these extensions would help clarify whether the limited contribution of county healthcare resources reflects their true relationship with preventive-care use or the limits of how access is measured at the county level.

7.2 Unsupervised Learning Findings

The unsupervised analysis showed that healthcare-resource availability and social vulnerability represent related but partly independent dimensions of county context. Five interpretable K-Means profiles emerged, separating provider-rich urban counties, resource-scarce but relatively low-vulnerability counties, and multiple forms of social vulnerability. One of the most surprising findings was that the counties with the lowest provider and hospital supply were not necessarily the counties with the greatest social vulnerability or the lowest routine-checkup utilization. Instead, the two most vulnerable profiles reflected different types of disadvantage: one centered on uninsurance, limited English proficiency, crowding, and a younger population, while the other was characterized more by disability, poverty, rurality, and FQHC reliance.

A central challenge was that the data did not contain sharply separated or objectively defined county types. Silhouette scores were low overall, and K-Means and GMM showed only moderate agreement in their assignments. Rather than treating this as evidence that clustering had failed, we interpreted the clusters as recognizable profiles along a broader continuum of county characteristics. We addressed this uncertainty by comparing two clustering methods with different assumptions, examining stability across random initializations, using PCA to assess geometric separation, and conducting sensitivity analyses with alternative feature weighting and HPSA indicators. K-Means and GMM recovered several similar broad structures but divided high-vulnerability counties differently, while the sensitivity analyses showed that finer cluster boundaries were particularly dependent on feature representation. These results make the broad patterns more credible while suggesting that individual cluster assignments and vulnerability subtypes should be interpreted cautiously.

The external comparison with routine-checkup prevalence provided another important lesson. Although utilization differed only modestly across profiles and the distributions overlapped substantially, the lowest median checkup prevalence occurred consistently in profiles characterized by high uninsurance and related access barriers rather than in the profile defined primarily by low provider supply. This suggests that examining healthcare resources alone would have missed important differences among counties and supports a multidimensional interpretation of healthcare access. Because routine-checkup prevalence was excluded from cluster construction and introduced only afterward, this pattern provides a useful descriptive comparison without implying that cluster membership or any individual feature causes differences in preventive-care utilization.

With additional time and resources, the analysis could be extended using spatial or multilevel methods that explicitly account for relationships among neighboring counties and counties within the same state. Additional insurance, healthcare-system, and policy measures could also help distinguish why counties with similar provider supply exhibit different utilization patterns. Longitudinal data would allow the stability of the identified profiles to be examined over time, while more precise historical HPSA data could improve the representation of provider shortage. These extensions would help determine which aspects of the observed county structure are persistent and which are sensitive to geography, measurement choices, or changing healthcare conditions.

7.3 Integrating the Two Analyses

The supervised and unsupervised analyses point to the same broader conclusion: **healthcare-resource availability alone does not fully explain preventive healthcare utilization**.

In the supervised models, geographic context was more predictive than individual provider measures, and adding broader social-vulnerability characteristics substantially improved prediction. In the clustering analysis, counties with the lowest provider supply did not necessarily have the greatest social vulnerability or lowest routine-checkup prevalence. Instead, lower utilization was most apparent in profiles characterized by barriers such as uninsurance.

Together, the results suggest that healthcare access is multidimensional. Workforce and infrastructure remain relevant, but utilization also reflects social, geographic, insurance, policy, and healthcare-system context. These analyses identify associations and predictive patterns rather than causal effects.

7.4 Limitations and Future Work

Several limitations affect interpretation. PLACES outcomes are model-based small-area estimates rather than direct county measurements, and their construction incorporates demographic information. Source years are not perfectly aligned, although the analysis targets approximately 2022 conditions.

The HPSA historical snapshot was reconstructed from current HRSA extracts rather than obtained from an archived 2022 dataset, so historical designation status is approximate and current scores may not equal 2022 values. Connecticut's geographic transition also produces structured missingness in historical AHRF measures.

Counties are not statistically independent; nearby counties and counties within the same state may share healthcare markets, policies, and infrastructure. State-grouped validation partially addresses this concern but is not a full spatial model. Healthcare resources are also not randomly distributed, preventing causal interpretation.

Finally, all results are ecological and cannot be assumed to describe individual residents. Future work could incorporate explicit spatial or multilevel models, additional insurance and healthcare-system measures, longitudinal data, and more precise historical HPSA information.

8. Ethical Considerations

Although the project uses public county-level data and no personally identifiable information, both the supervised and unsupervised analyses raise important ethical concerns related to interpretation, fairness, and downstream use.

Part A — Supervised Learning. The supervised model predicts county-level routine-checkup prevalence from healthcare-resource and access characteristics. A key ethical risk is that predictive associations could be misinterpreted as causal relationships. For example, a feature with high predictive importance should not be interpreted as evidence that changing that feature would necessarily change routine-checkup utilization. Model performance may also vary across geographic or demographic contexts that are not fully represented by the available predictors, particularly because counties differ in policy environments, healthcare systems, population size, and other unmeasured characteristics. Predictions should therefore not be used alone to rank communities, determine eligibility, or allocate healthcare resources. Appropriate use requires treating the model as a descriptive and decision-support tool, examining uncertainty and failure cases, and combining model output with local knowledge and additional evidence.

Part B — Unsupervised Learning. The clustering analysis introduces different ethical concerns because it assigns counties to descriptive healthcare-access and social-vulnerability profiles. Labels such as “high vulnerability,” “high hardship,” or “resource scarce” could stigmatize communities if interpreted as fixed or inherent characteristics rather than summaries of measured county-level conditions. Cluster boundaries are also sensitive to modeling choices, and counties within the same cluster may still differ substantially. These profiles should therefore not be treated as definitive classifications of communities or used by themselves to target interventions. In addition, county averages can conceal substantial within-county inequality and should not be generalized to individual residents.

Across both analyses, the results are ecological and observational rather than causal. Predictions, feature importance, and cluster membership should be interpreted as patterns in county-level data rather than evidence about individual people or proof that particular healthcare resources cause differences in utilization. The analyses are therefore best used to support further investigation and planning alongside community input, local expertise, and more detailed evidence.

9. Statement of Work

Table 6. Statement of Work and Team Contributions

Team Member	Primary Contributions	Shared Contributions
Samantha Salazar	Data acquisition and audit; geographic harmonization; HPSA reconstruction; multi-source integration; cleaning and feature engineering; analytical dataset and feature manifest; data/methodological documentation; policy and health-equity framing	Methodology review; interpretation; report writing; reproducibility and final review
Sara Jahanian	Supervised EDA and modeling; Elastic Net, SVR, and Gradient Boosting; model tuning and evaluation; sensitivity analysis; feature importance and ablation; failure analysis; supervised visualizations	Methodology review; interpretation; report writing; reproducibility and final review
Gina Sheesley	Unsupervised EDA and modeling; K-means and GMM; clustering evaluation and stability analysis; sensitivity analysis; cluster interpretation; geospatial and unsupervised visualizations; comparison with routine-checkup utilization	Methodology review; interpretation; report writing; reproducibility and final review

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Appendix A. Data, Feature Definitions, and Data Quality

Table A1. Final analytical features used in the supervised and unsupervised analyses

Healthcare-resource variables were derived primarily from HRSA Area Health Resources Files (AHRF); shortage indicators were derived from HRSA Health Professional Shortage Area (HPSA) records; social-vulnerability indicators came from the 2022 CDC/ATSDR Social Vulnerability Index (SVI); and the supervised outcome came from CDC PLACES. Variables identified as sensitivity features were evaluated as alternative or supplemental representations rather than included in every primary model.

Feature	Definition	Source	Analytical Use
routine_checkup_pct	Age-adjusted prevalence (%) of adults reporting a routine medical checkup within the past year	CDC PLACES	Supervised outcome; post-clustering descriptive comparison only
primary_care_physicians_per_100k	Primary-care physicians per 100,000 population	AHRF	Supervised primary; unsupervised primary
nurse_practitioners_per_100k	Nurse practitioners per 100,000 population	AHRF	Supervised primary; unsupervised primary
physician_assistants_per_100k	Physician assistants per 100,000 population	AHRF	Supervised primary; unsupervised primary
hospitals_per_100k	Hospitals per 100,000 population	AHRF	Supervised primary; unsupervised primary
hospital_beds_per_1000	Hospital beds per 1,000 population	AHRF	Supervised primary; unsupervised primary
fqhcs_per_100k	Federally Qualified Health Centers per 100,000 population	AHRF	Supervised primary; unsupervised primary
rural_health_clinics_per_100k	Rural Health Clinics per 100,000 population	AHRF	Supervised primary; unsupervised primary

log_population_density	Natural-log transformation, log(1 + population density), of persons per square mile	AHRF-derived	Supervised primary; unsupervised primary
rucc_metro_status	Simplified Rural-Urban Continuum classification: RUCC 1–3 = Metro; RUCC 4–9 = Nonmetro	AHRF-derived	Supervised primary
pc_hpsa_any	Indicator equal to 1 if the county had at least one qualifying Primary Care HPSA designation in the approximate year-end 2022 reconstruction; 0 otherwise	HRSA HPSA	Supervised primary; unsupervised sensitivity/profiling
mh_hpsa_any	Indicator equal to 1 if the county had at least one qualifying Mental Health HPSA designation; 0 otherwise	HRSA HPSA	Unsupervised sensitivity/profiling
dh_hpsa_any	Indicator equal to 1 if the county had at least one qualifying Dental Health HPSA designation; 0 otherwise	HRSA HPSA	Unsupervised sensitivity/profiling
EP_POV150	Percentage of persons below 150% of the federal poverty level	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
EP_UNEMP	Percentage of the civilian population age 16 or older that is unemployed	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
EP_HBURD	Percentage of occupied housing units experiencing housing-cost burden	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
EP_NOHSDP	Percentage of persons age 25 or older without a high-school diploma	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
EP_UNINSUR	Percentage of the civilian noninstitutionalized population without health insurance	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity

EP_AGE65	Percentage of persons age 65 or older	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
EP_AGE17	Percentage of persons age 17 or younger	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
EP_DISABL	Percentage of the civilian noninstitutionalized population with a disability	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
EP_SNGPNT	Percentage of households with children that are single-parent households	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
EP_LIMENG	Percentage of persons age 5 or older who speak English less than well	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
EP_MINRTY	Percentage of the population belonging to racial or ethnic minority groups	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
EP_MUNIT	Percentage of housing units in structures containing 10 or more units	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
EP_MOBILE	Percentage of housing units that are mobile homes	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
EP_CROWD	Percentage of occupied housing units with more occupants than rooms	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
EP_NOVEH	Percentage of households with no vehicle available	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity

EP_GROUPQ	Percentage of persons living in group quarters	CDC/ATSDR SVI	Unsupervised primary; supervised SVI sensitivity
pc_hpsa_count	Number of unique qualifying Primary Care HPSA designations associated with the county in the approximate year-end 2022 reconstruction	HRSA HPSA-derived	Supervised sensitivity: alternative to pc_hpsa_any
pc_hpsa_max_score	Maximum recorded Primary Care HPSA score among qualifying county designations; structurally undefined when no qualifying designation exists	HRSA HPSA-derived	Supervised sensitivity
rural_urban_contnm_23	Detailed 2023 Rural-Urban Continuum Code (1–9)	AHRF	Supervised sensitivity: alternative rurality representation
popn_densty_per_squr_mi_20	Population density in persons per square mile	AHRF	Supervised sensitivity: alternative to log-transformed density

Notes. The seven provider/facility rate variables were derived from AHRF counts using population denominators to improve comparability across counties of different sizes. The primary supervised specification consisted of the seven resource-rate variables, log_population_density, rucc_metro_status, and pc_hpsa_any. The primary unsupervised feature matrix consisted of the seven resource-rate variables, log_population_density, and the 16 SVI indicators (24 continuous variables total). HPSA indicators were excluded from the primary clustering matrix and evaluated separately through sensitivity analysis and cluster profiling. The PLACES routine-checkup outcome was never used to construct clusters.

SVI -999 sentinel values were converted to missing values before analysis. HPSA indicators were based on an approximate reconstruction of designations active on December 31, 2022 using designation and withdrawal dates from the current HRSA extract; therefore, these variables should not be interpreted as an official archived 2022 HPSA snapshot.

Table A2. Data Source and Geographic Harmonization Audit

Data source/step	Geographic records	Harmonization result	Final treatment
CDC PLACES 2024 county release	3,144 county/county-equivalent records after excluding the U.S. aggregate	Defined the authoritative project geographic universe	Retained all 3,144 unique five-character county FIPS codes
CDC/ATSDR SVI	3,144 project	Exact FIPS coverage for the	Retained and joined

2022	county/county-equivalent FIPS	PLACES geographic universe	one-to-one
HRSA AHRF 2023–2024	3,240 unique geographic records	All 3,144 project FIPS were represented; 96 additional territorial, historical, or superseded geographies were outside the project universe	Retained only records matching the 3,144 PLACES FIPS
HRSA HPSA	Designation-level records rather than one row per county	Eligible records were reconstructed to approximate year-end 2022 status and aggregated to the county level	Joined to the same 3,144-county universe
Final integrated dataset	3,144 rows	Each incremental merge preserved 3,144 unique FIPS with no duplicate county observations	Final shared analytical dataset contains 3,144 county/county-equivalent observations

Note. Standardized five-character FIPS codes were used as the authoritative join key. County names were retained for interpretation and quality control but were not used as merge keys. Incremental joins were validated as one-to-one to ensure that counties were neither lost nor duplicated.

Table A3. Missing-Data and Geographic Exceptions

Issue	Type of missingness / data-quality concern	Treatment	Analytical implication
Connecticut planning regions	Structured geographic missingness in several historical AHRF healthcare-resource variables following Connecticut's transition from counties to planning regions	Retained in the shared dataset; supervised pipelines handled missing values. Nine planning regions were excluded from the primary clustering matrix because most relevant historical resource measures were unavailable	Supervised analysis retained national coverage; primary unsupervised sample was reduced from 3,144 to 3,135 observations
Alaska resource variables	Limited missingness for a small number of Alaska observations in the unsupervised feature set	Handled separately using documented imputation before clustering	Counties were retained rather than removed for limited missingness
SVI sentinel values	Encoded missingness represented numerically as `-999`	Converted to missing before analysis	Prevented sentinel codes from being interpreted as valid extreme values
Counties with no qualifying HPSA designation	Structural absence rather than missing data	HPSA presence = 0 and HPSA count = 0	Absence of a designation was represented explicitly
HPSA maximum score when no designation exists	Structural missingness	Not automatically interpreted as a score of zero during data auditing; score was treated separately from designation absence	Avoided equating “no designation” with a true HPSA score of zero
Shared analytical dataset	Mixed true, structured, and encoded missingness	No global imputation was applied before modeling	Missing-data handling remained method-specific and reduced the risk of

			inappropriate preprocessing assumptions
Supervised preprocessing	Potential leakage from preprocessing before cross-validation	Numeric imputation, scaling, transformations, and categorical encoding were performed inside model pipelines	Preprocessing parameters were estimated from training data/folds only
Unsupervised preprocessing	Clustering requires a complete standardized feature matrix	Missingness was resolved before clustering, followed by transformations and standardization	Produced a consistent 3,135-county primary clustering matrix

Note. Missing values were not treated uniformly because their meanings differed across sources. In particular, Connecticut gaps were interpreted as structured missingness rather than failed geographic joins or true zero resource values.

Table A4. HPSA Historical Reconstruction and County Aggregation Rules

Step	Final rule	Rationale / limitation
Historical reference date	Approximate designation status as of **December 31, 2022**	Aligns HPSA status as closely as possible with the 2022 PLACES outcome period
Active designation rule	`HPSA Designation Date` ≤ `2022-12-31` and `Withdrawn Date` is missing or after `2022-12-31`	Reconstructs whether a designation was plausibly active at year-end 2022
Included designation types	Geographic HPSA; High Needs Geographic HPSA; HPSA Population	Focuses on area- and population-level provider shortages
Excluded designation types	Facility-specific HPSAs	Facility-specific shortage status does not necessarily characterize the broader county population
Disciplines retained	Primary Care; Mental Health; Dental Health	Represents the major HPSA disciplines available in HRSA data
Deduplication key	`HPSA ID + county FIPS`	Prevents one designation with multiple component records from being counted repeatedly within the same county
County-level presence feature	`1` if ≥1 qualifying designation; otherwise `0`	Provides a simple and robust designation indicator
County-level count feature	Number of unique qualifying designations; `0` if none	Used primarily for sensitivity analysis
County-level maximum score	Maximum recorded HPSA score among qualifying designations	Used primarily for sensitivity analysis because current scores may not equal their exact historical 2022 values
Shortage/FTE fields	Not summed across designation records	These values may repeat across county components and could be inflated by naïve aggregation

Primary analytical emphasis	Designation presence	Considered more defensible than count or current score for the approximate historical reconstruction
Historical-data caveat	Reconstruction is based on a current HRSA extract	It is **not** an official archived 2022 HPSA snapshot

Note. The HPSA reconstruction should be interpreted as an approximate historical representation. A designation may have been active during 2022 while its score in the current extract was subsequently updated, so score-based results were treated as sensitivity analyses rather than the primary shortage specification.

Table A5. Feature Engineering and Analytical Representation

Raw/source measure	Engineered or retained representation	Transformation / coding	Analytical purpose
Primary-care physician count	`primary_care_physicians_per_100k`	Count standardized per 100,000 population	Provider-supply measure
Nurse practitioner count	`nurse_practitioners_per_100k`	Count standardized per 100,000 population	Provider-supply measure
Physician assistant count	`physician_assistants_per_100k`	Count standardized per 100,000 population	Provider-supply measure
Hospital count	`hospitals_per_100k`	Count standardized per 100,000 population	Healthcare-infrastructure measure
Hospital-bed count	`hospital_beds_per_1000`	Beds standardized per 1,000 population	Healthcare-capacity measure
FQHC count	`fqhcs_per_100k`	Count standardized per 100,000 population	Safety-net / infrastructure measure
Rural Health Clinic count	`rural_health_clinics_per_100k`	Count standardized per 100,000 population	Rural healthcare-infrastructure measure
Population density	`log_population_density`	Log transformation of population density because of strong right skew	Geographic/access context; reduces influence of extreme urban density
Rural-Urban Continuum Code	`rucc_metro_status`	Detailed RUCC collapsed to metro/nonmetro for the primary supervised model	Provides an interpretable rurality indicator; detailed RUCC retained for sensitivity analysis
Primary-care HPSA records	`pc_hpsa_any`	Binary presence of ≥1 qualifying reconstructed designation	Primary provider-shortage feature
Primary-care HPSA records	`pc_hpsa_count`, `pc_hpsa_max_score`	Unique designation count and maximum score	Supervised sensitivity features
Mental-health and dental HPSA records	`mh_hpsa_any`, `dh_hpsa_any`	Binary designation-presence variables	Unsupervised profiling and sensitivity analysis
SVI percentage indicators	16 underlying `EP_*` indicators	`-999` converted to missing; component indicators retained rather	Primary social-vulnerability representation for clustering;

		than combined with derived theme/overall rankings	supervised sensitivity analysis
PLACES routine-checkup estimate	`routine_checkup_pct`	2022 age-adjusted prevalence retained as the outcome	Supervised target; excluded from cluster construction and used only after clustering for descriptive comparison
Geographic identifiers	FIPS, state, county/county-equivalent name	FIPS standardized to five characters	Joining, reporting, mapping, and interpretation only; not used as ML predictors
Continuous clustering features	8 AHRF resource/context measures + 16 SVI indicators	Skewed variables transformed where appropriate; all primary continuous clustering features standardized before fitting	Ensures variables on different scales contribute comparably to K-Means and GMM

Note. Population-standardized rates were preferred to raw provider/facility counts because raw counts are strongly influenced by county population size. Raw source measures were retained for traceability. The primary supervised specification ultimately used 10 predictors, while the primary unsupervised representation used 24 continuous healthcare-resource/context and SVI features.

Appendix B. Supplemental Supervised Learning Evidence

Table B1. Hyperparameter Search Spaces and Selected Model Configurations

Model	Hyperparameters evaluated	Selected configuration	Best tuning-fold MAE
DummyRegressor	No tuning; median prediction baseline	`strategy = "median"`	—
Elastic Net	α: 15 log-spaced values from 0.001 to 3.162; `l1_ratio`: 0.1, 0.3, 0.5, 0.7, 0.9, 1.0; rate representation: raw or `log1p`	α = 0.01; `l1_ratio` = 1.0; rate features = log1p	2.334
SVR	C: 0.3, 1, 3, 10, 30; ε: 0.1, 0.5, 1.0; γ: `scale`, 0.01, 0.03, 0.1; rate representation: raw or `log1p`	RBF kernel; C = 10; ε = 0.5; γ = scale; rate features = log1p	2.116
Gradient Boosting	Learning rate: 0.03, 0.10; trees: 300, 600; max depth: 2, 3, 4; minimum leaf size: 20; subsample: 0.8; loss: squared error or Huber	Learning rate = 0.03; trees = 300; max depth = 4; minimum leaf size = 20; subsample = 0.8; loss = Huber	2.106

Note. Hyperparameters were selected using cross-validation within the training data with MAE as the refit criterion. The reserved holdout test set was not used for model tuning or selection.

Table B2. Complete Cross-Validation Performance for the Primary Supervised Specification

Model	CV scheme	MAE mean ± SD	RMSE mean ± SD	R^2 mean ± SD	Training MAE	Validation–training MAE gap
Dummy	Random	2.813 ± 0.064	3.631 ± 0.063	−0.027 ± 0.009	2.813	0.000
Elastic Net	Random	2.329 ± 0.052	2.999 ± 0.078	0.299 ± 0.034	2.317	0.012
SVR	**Random**	**2.105 ± 0.027**	**2.756 ± 0.035**	**0.408 ± 0.018**	1.916	0.189
Gradient Boosting	Random	2.116 ± 0.064	2.776 ± 0.054	0.399 ± 0.019	1.687	0.429
Dummy	State-grouped	3.056 ± 0.744	3.794 ± 0.884	−0.698 ± 0.766	2.783	0.273
Elastic Net	State-grouped	2.569 ± 0.448	3.226 ± 0.526	−0.248 ± 0.484	2.292	0.278
SVR	**State-grouped**	**2.455 ± 0.562**	3.081 ± 0.639	−0.157 ± 0.538	1.855	0.600
Gradient Boosting	State-grouped	2.461 ± 0.597	**3.079 ± 0.649**	−0.170 ± 0.577	1.653	0.808

Note. Results summarize five cross-validation folds. MAE was the primary model-selection metric. SVR was selected because it had the lowest mean random-CV MAE. Gradient Boosting was nearly tied but showed a larger training-validation gap. State-grouped CV entire states out of training and therefore **evaluates a more difficult geographic generalization setting.**

Figure B1. Cross-validated model comparison

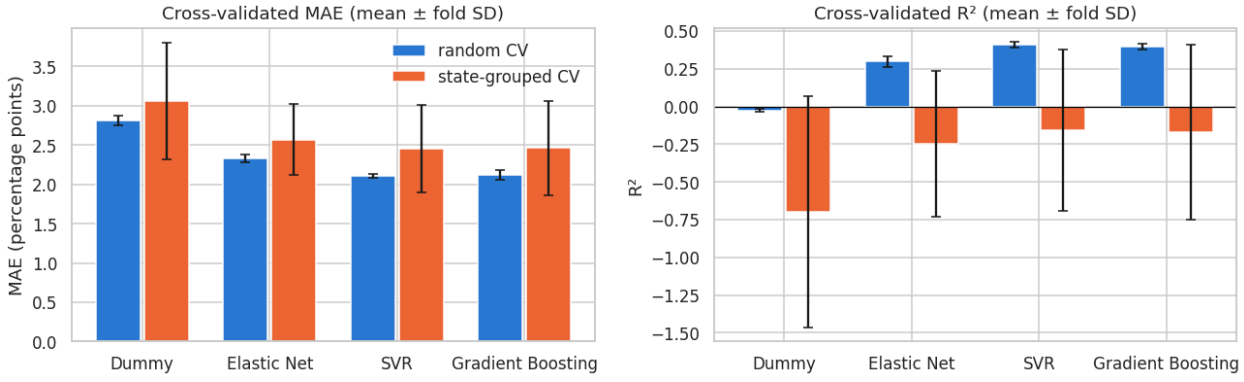


Figure B2. Permutation importance across all three model families

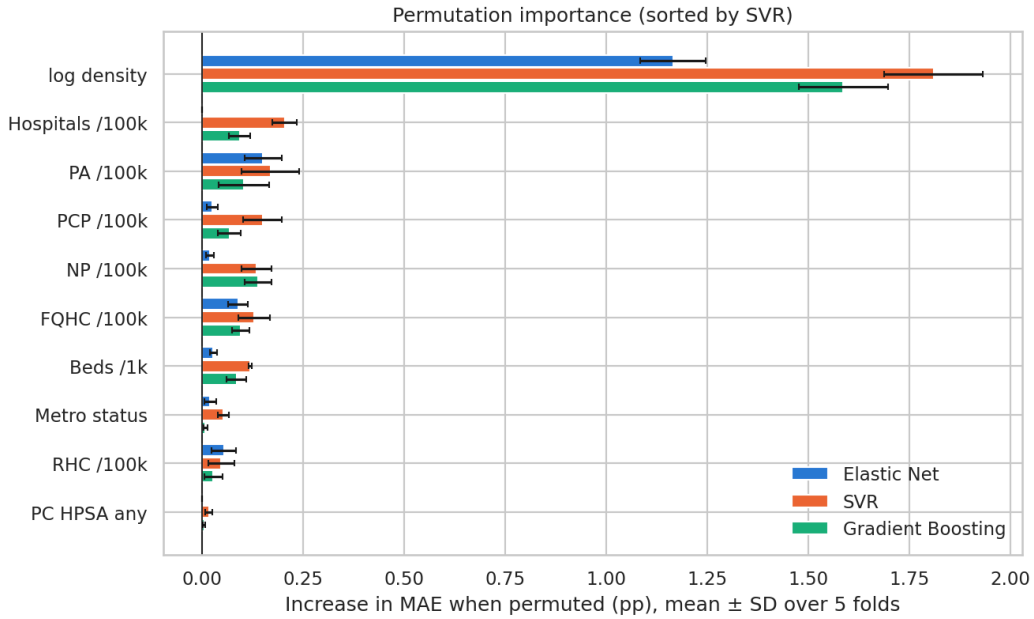


Table B3. Complete Feature-Group Ablation Results for the Selected SVR

Feature-group specification	CV scheme	MAE	ΔMAE vs. full SVR	Interpretation
Full SVR	Random	**2.105**	—	Primary specification
Drop geographic/access context	Random	2.441	**+0.336**	Largest loss in predictive performance
Drop healthcare infrastructure	Random	2.229	+0.124	Infrastructure contributes unique predictive information
Drop provider supply	Random	2.158	+0.053	Provider supply contributes a smaller unique gain
Drop provider shortage	Random	2.105	~0.000	HPSA presence adds essentially no unique random-CV benefit
Geographic/access context only	Random	2.359	+0.254	Geography alone captures substantial predictive signal
Healthcare infrastructure only	Random	2.584	+0.479	Insufficient as a stand-alone feature domain
Provider supply only	Random	2.580	+0.475	Insufficient as a stand-alone feature domain
Provider shortage only	Random	2.810	+0.705	Performs close to the constant baseline
Full SVR	State-grouped	**2.455**	—	Primary specification
Drop geographic/access context	State-grouped	2.856	**+0.401**	Largest geographic-generalization loss

Drop healthcare infrastructure	State-grouped	2.535	+0.080	Smaller but positive contribution
Drop provider supply	State-grouped	2.478	+0.023	Limited unique contribution
Drop provider shortage	State-grouped	2.437	−0.018	No evidence of improved performance from retaining HPSA presence
Geographic/access context only	State-grouped	2.603	+0.148	Strongest stand-alone domain
Healthcare infrastructure only	State-grouped	2.920	+0.465	Weak stand-alone performance
Provider supply only	State-grouped	2.889	+0.434	Weak stand-alone performance
Provider shortage only	State-grouped	3.082	+0.627	Weakest stand-alone domain

Note. "Drop" specifications remove one conceptual feature group from the full SVR, while "only" specifications retain only that group. Positive ΔMAE indicates worse performance relative to the full primary SVR.

Figure B3. Full feature-group ablation

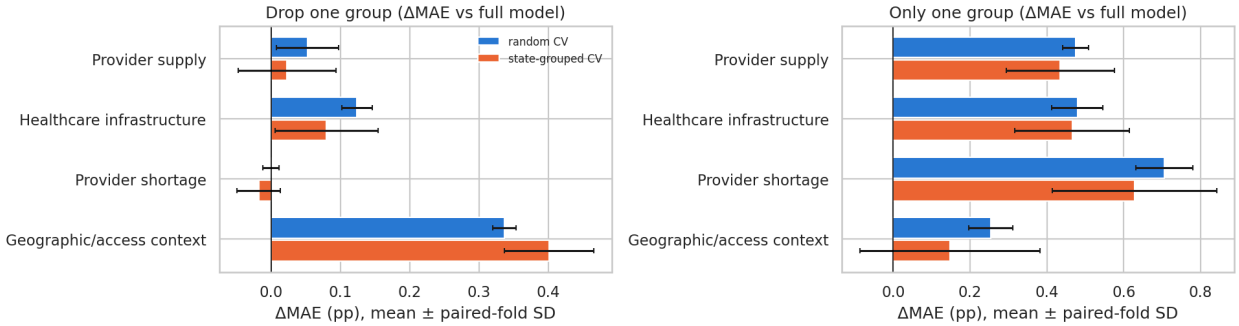


Figure B4. Out-of-fold model predictions

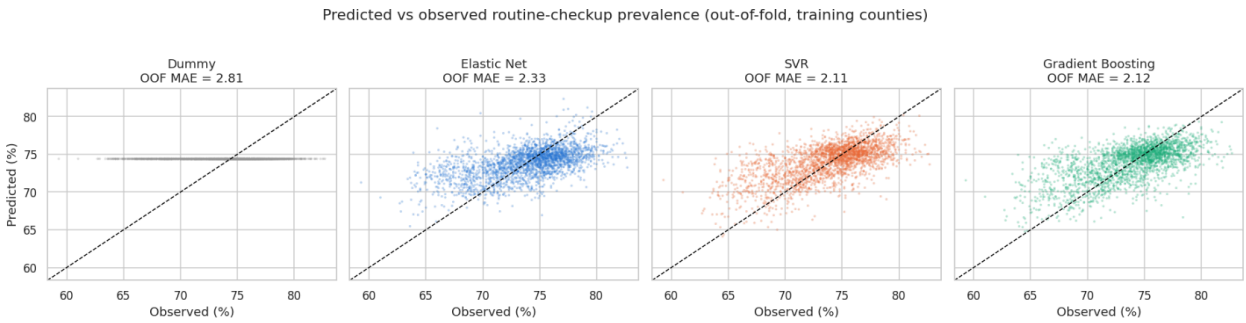


Table B4. Complete Supervised Sensitivity Analysis

Specification	Model	CV scheme	Features	MAE	R^2	ΔMAE vs. primary	Paired-fold Δ SD	Folds better
Primary	SVR	Random	10	2.105	0.408	0.000	0.000	—
Primary	SVR	State-grouped	10	2.455	−0.157	0.000	0.000	—
Primary	Elastic Net	Random	10	2.329	0.299	0.000	0.000	—
Primary	Elastic Net	State-grouped	10	2.569	−0.248	0.000	0.000	—
HPSA count	SVR	Random	10	2.094	0.413	−0.011	0.019	4/5
HPSA count	SVR	State-grouped	10	2.476	−0.179	+0.021	0.017	1/5
HPSA count	Elastic Net	Random	10	2.321	0.306	−0.009	0.011	3/5
HPSA count	Elastic Net	State-grouped	10	2.559	−0.235	−0.011	0.038	4/5
HPSA score	SVR	Random	11	2.104	0.411	−0.001	0.016	4/5
HPSA score	SVR	State-grouped	11	2.500	−0.190	+0.045	0.030	1/5
HPSA score	Elastic Net	Random	11	2.324	0.303	−0.005	0.008	4/5

HPSA score	Elastic Net	State-grouped	11	2.586	−0.261	+0.016	0.021	2/5
Detailed rurality	SVR	Random	10	2.100	0.409	−0.005	0.008	3/5
Detailed rurality	SVR	State-grouped	10	2.452	−0.157	−0.003	0.025	4/5
Detailed rurality	Elastic Net	Random	10	2.306	0.310	−0.023	0.006	5/5
Detailed rurality	Elastic Net	State-grouped	10	2.554	−0.238	−0.015	0.032	4/5
Raw density	SVR	Random	10	2.418	0.231	**+0.313**	0.020	0/5
Raw density	SVR	State-grouped	10	2.819	−0.462	**+0.364**	0.070	0/5
Raw density	Elastic Net	Random	10	2.602	0.136	+0.273	0.029	0/5
Raw density	Elastic Net	State-grouped	10	2.904	−0.540	+0.335	0.072	0/5
Add 16 SVI indicators	SVR	Random	26	**1.624**	**0.621**	**−0.482**	0.045	5/5
Add 16 SVI indicators	SVR	State-grouped	26	**2.124**	0.051	**−0.331**	0.134	5/5
Add 16 SVI indicators	Elastic Net	Random	26	1.904	0.514	−0.425	0.023	5/5

Add 16 SVI indicators	Elastic Net	State-grouped	26	2.156	0.058	−0.413	0.192	5/5

Note. Negative ΔMAE indicates improvement relative to the corresponding primary specification. HPSA and rurality alternatives produced only small changes. Using raw rather than log-transformed density consistently worsened performance. Adding SVI produced the largest improvement, but the SVI specification remains a sensitivity analysis because of conceptual overlap with demographic and socioeconomic inputs used in PLACES estimation.

Figure B5. Supervised sensitivity specifications

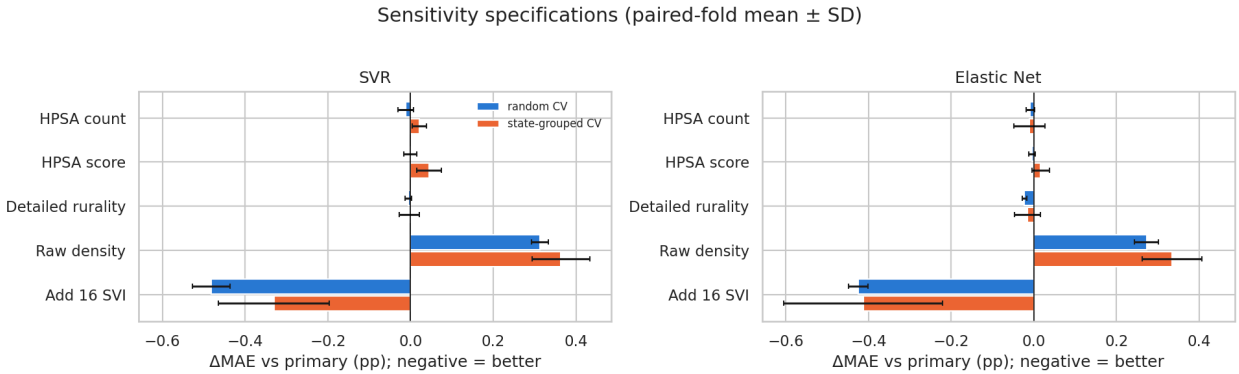


Figure B6. SVI-expanded SVR permutation importance

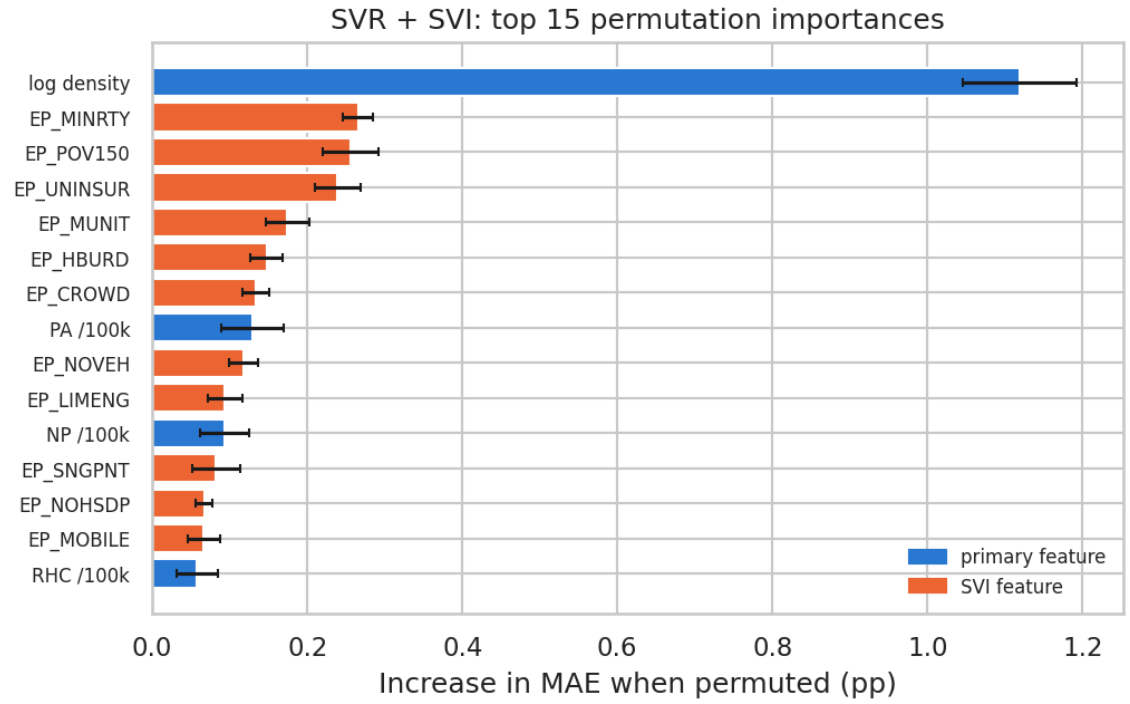


Table B5. Final Holdout Test Performance

Model	Test MAE	Test RMSE	Test R^2	Random-CV MAE	State-grouped CV MAE
SVR	2.053	2.696	0.427	2.105	2.455

Holdout MAE = **2.053 percentage points** with a **95% bootstrap CI of 1.921–2.186**. The mean improvement over the Dummy baseline was **0.768 percentage points**, with a **95% bootstrap CI of 0.617–0.904**.

Note. SVR was selected using cross-validation before the reserved holdout test set was evaluated. The holdout set was used only for final evaluation of the selected model and was not used for model selection or hyperparameter tuning.

Figure B7. Final holdout evaluation

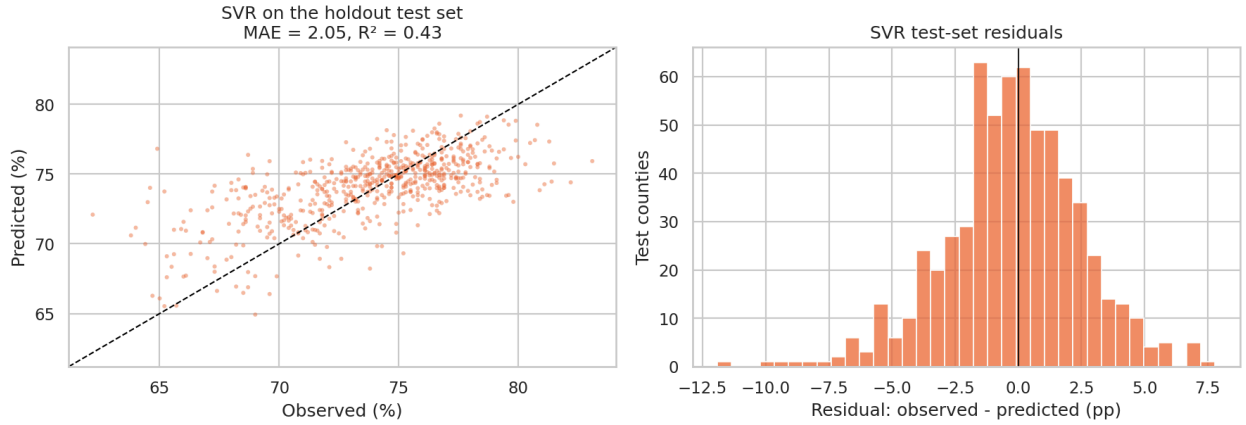


Figure B8. SVR partial-dependence relationships

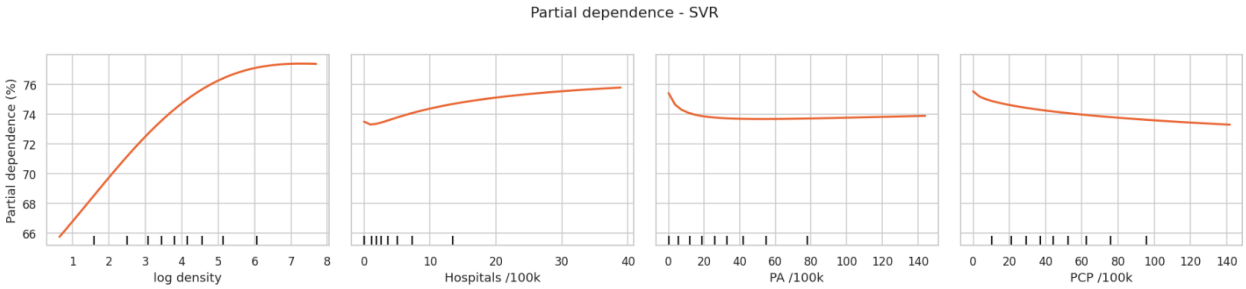


Table B6. Fifteen Largest Absolute Out-of-Fold SVR Prediction Errors

FIPS	State	County	Observed (%)	Predicted (%)	Residual	Population	Metro status
15005	HI	Kalawao	59.3	71.60	**−12.30**	82	Metro
53069	WA	Wahkiakum	63.9	74.45	**−10.55**	4,688	Nonmetro
35061	NM	Valencia	64.9	75.07	**−10.17**	78,080	Metro
35001	NM	Bernalillo	65.3	75.30	**−10.00**	672,508	Metro
08031	CO	Denver	66.0	76.00	**−10.00**	713,252	Metro
02275	AK	Wrangell	63.8	73.68	**−9.88**	2,070	Nonmetro
23021	ME	Piscataquis	78.9	69.05	**+9.85**	17,417	Nonmetro
08047	CO	Gilpin	66.9	76.47	**−9.57**	5,891	Metro
27109	MN	Olmsted	67.2	76.35	**−9.15**	164,020	Metro
28021	MS	Claiborne	82.7	73.66	**+9.04**	8,805	Nonmetro
23025	ME	Somerset	78.4	69.43	**+8.97**	51,098	Nonmetro
48301	TX	Loving	61.0	69.96	**−8.96**	51	Nonmetro

53055	WA	San Juan	66.0	74.86	**−8.86**	18,662	Nonmetro
53015	WA	Cowlitz	64.3	72.92	**−8.62**	111,956	Metro
53053	WA	Pierce	66.6	75.09	**−8.49**	927,380	Metro

Note. Residual = observed − predicted, so negative values indicate overprediction and positive values indicate underprediction. These are out-of-fold predictions: each county was predicted by an SVR that was fitted without that county in its training fold. The largest errors reinforce the broader tendency for predictions to be compressed toward the center of the observed outcome distribution.

Figure B9. State-level geographic bias

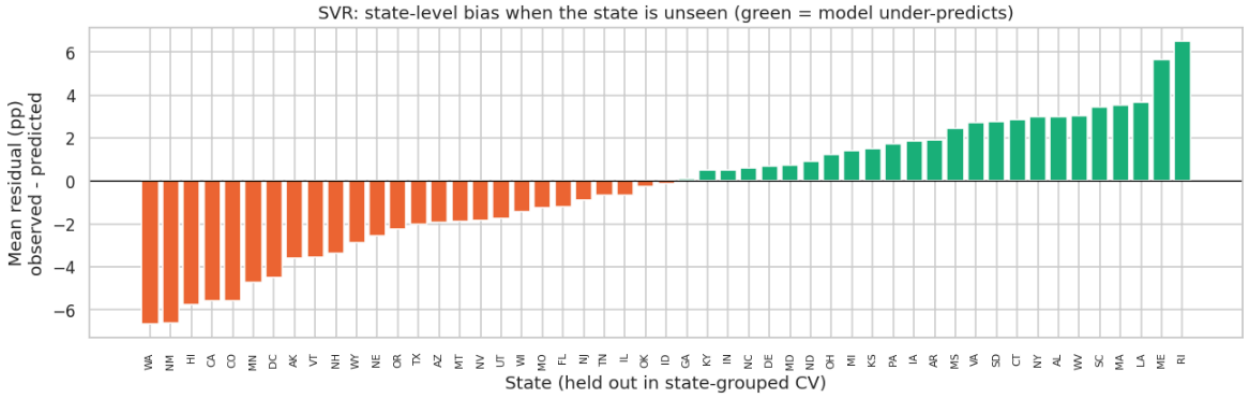
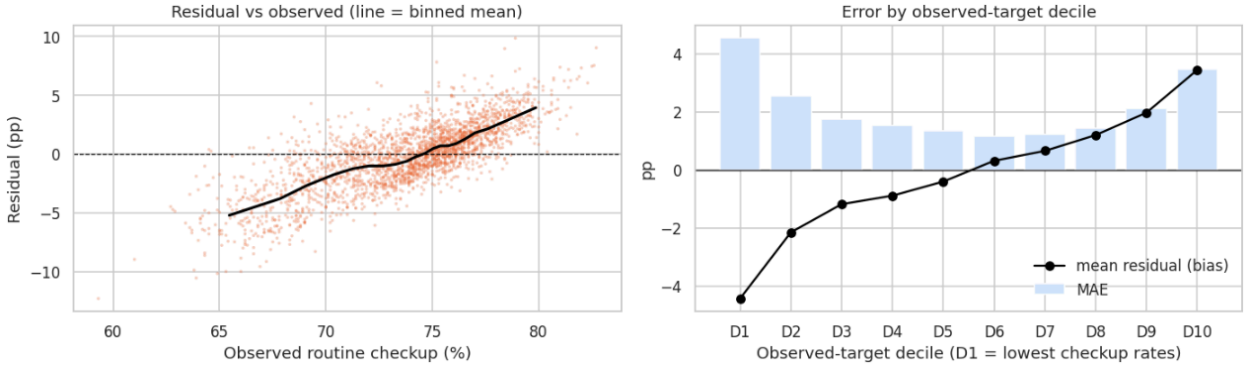


Figure B10. Failure pattern by observed outcome



Appendix C. Supplementary Unsupervised Learning Evidence

The following tables provide additional detail supporting the unsupervised learning analysis presented in Section 6. The primary clustering analysis used 3,135 counties/county equivalents and 24 standardized healthcare-resource, geographic/access, and social-vulnerability features. Routine-checkup prevalence was excluded from cluster formation and examined only after the cluster assignments were finalized.

Figure C1. K-Means Elbow Diagnostic Across Candidate Cluster Counts

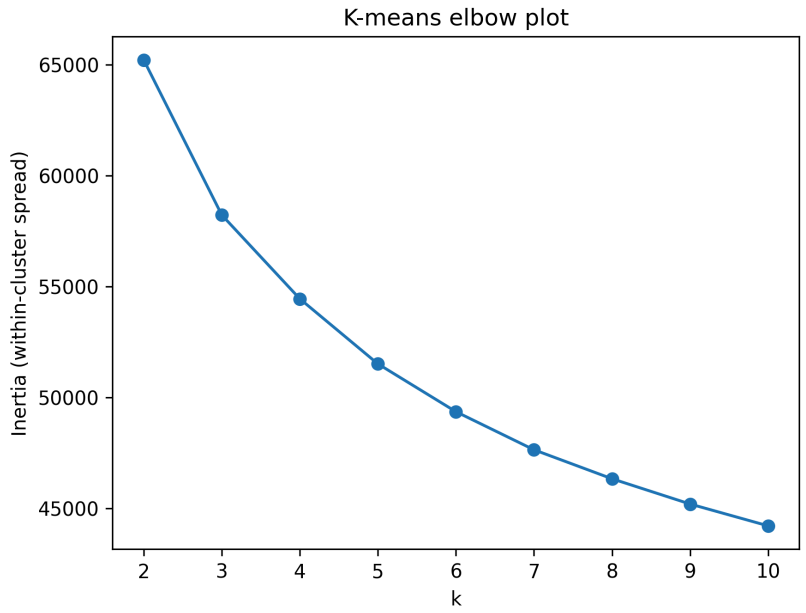


Table C1. Selected HPSA Designation Evidence by Cluster/Profile

County group	Primary-care HPSA	Mental-health HPSA	Dental HPSA	Interpretation
Full clustering sample	85.8%	89.0%	71.6%	HPSA designations are common nationally
Urban/provider-rich profile	~70–77%	~70–77%	~56–60%	Lowest designation rates among the major profiles
K-Means Cluster 2: resource-scarce, low vulnerability	~86%	—	~65%	Despite the lowest provider/hospital supply, designation rates are near the sample average
K-Means Cluster 4: rural high-hardship	~97%	~96%	90.8%	Very high shortage-designation prevalence accompanies high social vulnerability

Note. HPSA variables were not included in the primary continuous clustering matrix. They were used afterward for cluster profiling and as alternative features in sensitivity analyses. The pattern suggests that administrative shortage designation is related to, but not interchangeable with, observed provider supply.

Table C2. Selected Routine-Checkup Utilization by Cluster/Profile

Comparison	Median routine-checkup prevalence	Interpretation
Overall unsupervised analysis sample	74.4%	Reference median
K-Means high-uninsurance/access-barrier profile	71.6%	Lowest K-Means median; characterized more strongly by insurance and social-access barriers than provider scarcity
GMM high-uninsurance profile	71.5%	Lowest GMM median, reproducing the same broad association under a different clustering method
K-Means rural high-hardship profile	75.7%	Highest reported cluster median despite elevated poverty and disability
K-Means provider-scarce profile	Near overall median	Low provider supply alone was not associated with the lowest routine-checkup prevalence

Note. Routine-checkup prevalence was not used to construct the clusters. It was reintroduced only after clustering as an external descriptive comparison. Cluster distributions overlapped substantially; these results should therefore be interpreted descriptively rather than causally.